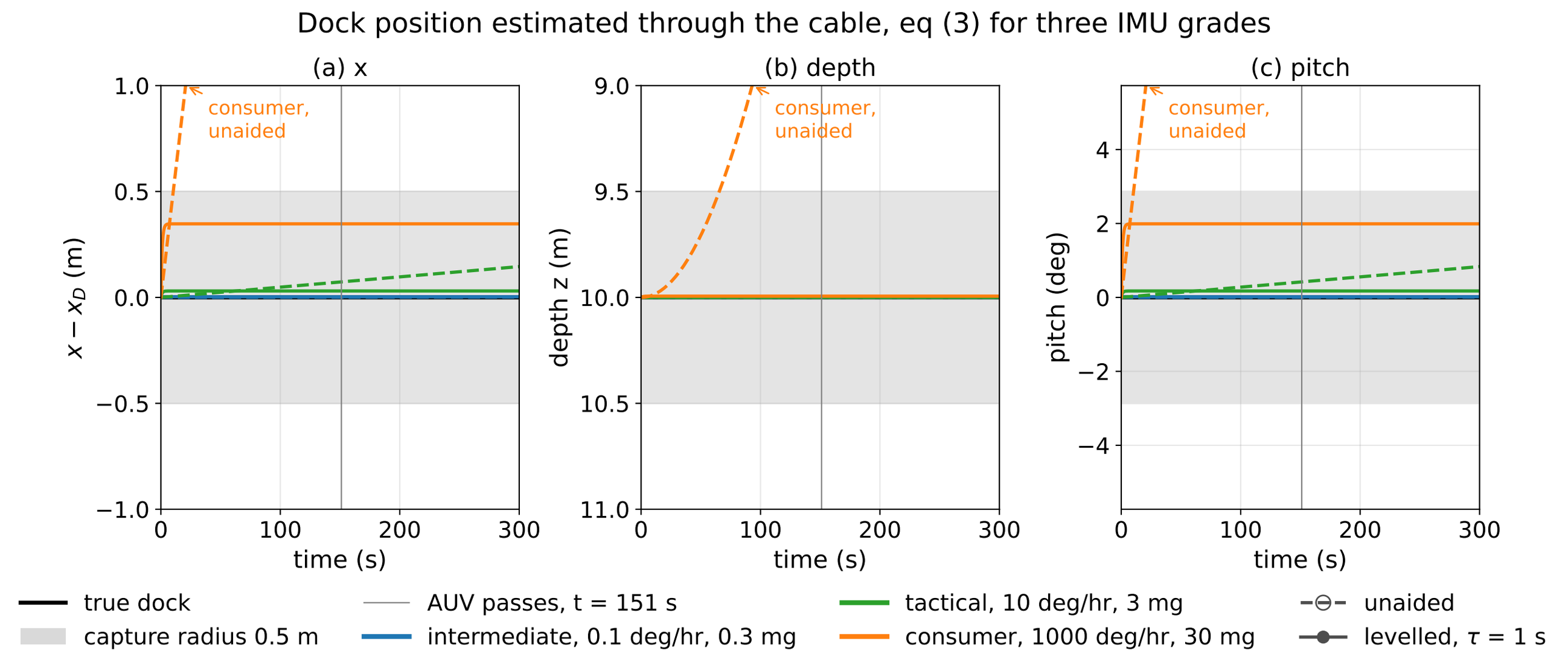
Dock position estimated through the cable, eq (3) for three IMU grades
(a) x
(b) depth
(c) pitch
consumer, unaided
$x - x_D$ (m)
depth z (m)
pitch (deg)
time (s)
true dock
capture radius 0.5 m
AUV passes, t = 151 s
intermediate, 0.1 deg/hr, 0.3 mg
tactical, 10 deg/hr, 3 mg
consumer, 1000 deg/hr, 30 mg
unaided
levelled, τ = 1 s